A Graph-Augmented Bayesian Simulation Engine for SME Marketing Intelligence in Emerging Markets

Technical Whitepaper

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Abstract. Bangladesh's rapidly digitising SME sector now invests significant capital across Meta Ads, Google, and TikTok, yet no predictive marketing intelligence service exists for this market. Local brands operate exclusively with retrospective platform dashboards, allocating budgets through intuition rather than mathematical simulation. Enterprise solutions from global providers are both economically inaccessible and architecturally unsuitable—built on Multi-Touch Attribution models that have been rendered unreliable by walled-garden ecosystems and privacy regulation. This paper presents the **Brand Simulation Engine**: the first causally rigorous, pre-launch campaign simulation system designed specifically for Bangladeshi SMEs. The architecture synthesises Bayesian

Marketing Mix Modelling (MMM) with Adstock and Hill function transformations, Agent-Based Modelling (ABM), Markov Chain attribution, and NSGA-II Genetic Algorithm optimisation within a Graph-Retrieval Augmented Generation (GraphRAG) backend powered by Neo4j. Real-time competitor intelligence is ingested autonomously via Firecrawl and Crawl4AI, while Bangla-native NLP is delivered through `csebuetnlp/banglabert` and `BAAI/bge-m3` embeddings. The platform is deployed on Vercel using Next.js 15 with aggressive low-bandwidth edge optimisations and an offline-capable local LLM fallback via Ollama (Qwen3-8B), ensuring accessibility on the 2G/3G infrastructure prevalent across rural Bangladesh. All simulation outputs are grounded by SHAP-guaranteed deterministic explanations before synthesis into a natural-language executive report.

Keywords: Brand Simulation Engine, Marketing Mix Modelling, Bayesian Inference, GraphRAG, Agent-Based Modelling, Genetic Algorithms, SME Analytics, Bangla NLP, Bangladesh, Emerging Markets

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1 Introduction

1.1 The Market Reality in Bangladesh

Bangladesh is in the middle of a profound digital transition. With a population exceeding 170 million, a rapidly expanding smartphone base, and one of the fastest-growing digital advertising markets in South Asia, the country's SME sector now routinely deploys significant capital across Meta Ads, Google, and TikTok. Yet despite this substantial and growing investment, the brands making these decisions are doing so almost entirely blind.

There is, at present, **no predictive marketing intelligence service of any kind available to Bangladeshi SMEs.** The tools accessible to local marketers are exclusively *retrospective*: platform-native dashboards (Meta Ads Manager, Google Analytics) that report what has already happened—impressions served, clicks recorded, conversions logged. They offer no capacity to answer the questions that actually determine strategic outcomes: *"What will happen if I increase my Meta budget by 30%?" "Is my Google spend producing incremental sales, or am I capturing demand that would have converted organically regardless?" "At what budget level does my TV campaign stop delivering real returns?"*

The consequence is that marketing budget allocation in Bangladesh—across thousands of SMEs collectively spending hundreds of crore taka per year—is governed not by mathematical evidence but by intuition, platform salespeople, and month-over-month comparisons of vanity metrics. Agencies optimise for clicks. Clients optimise for impressions. Neither party can prove, with any statistical rigour, that the advertising actually caused the sales that followed.

1.2 Why Global Tools Do Not Solve This Problem

A natural question is whether established Western analytics platforms could simply be adopted. The answer is no, for reasons both economic and structural.

Enterprise-grade solutions such as Google Meridian, Nielsen Marketing Cloud, and Analytic Partners are architected for large multinationals with dedicated data science teams, multi-year historical datasets, and six-figure annual licensing budgets. They are fundamentally inaccessible to SMEs operating on margins common in the Bangladeshi market.

Beyond cost, even if these tools were accessible, the dominant paradigm they built upon—**Multi-Touch Attribution (MTA)**—has structural failures that make it particularly inadequate for this context. MTA models track individual user journeys via tracking pixels and third-party cookies. But the rise of closed "walled-garden" ecosystems (Meta, Google) and the global enforcement of privacy legislation have severely fractured user-level tracking [1]. Data that MTA depends on is either unavailable, incomplete, or misrepresentative. More fundamentally, MTA is *descriptive rather than causal*—it can report which channels a user touched before converting, but it cannot prove that removing any one of those channels would have reduced conversions. It is a historical ledger, not a simulation engine.

1.3 The Four Structural Blind Spots

Beyond the MTA problem, any attempt to build a predictive system for this market must confront four compounding analytical failures inherent to conventional forecasting:

(i) **Cold-Start Failure.** Regression models produce no meaningful prediction for entirely new products or campaigns with no historical interaction data— a common scenario for SMEs entering new channels or launching new product lines.

(ii) **Linear Scalability Fallacy.** Naïve models assume proportional returns: if $1,000 drives 100 sales, $100,000 will drive 10,000. In reality, diminishing returns set in far earlier, and conventional tools provide no mechanism to identify that threshold.

(iii) **Temporal Lag Neglect.** A consumer who sees an ad today may not convert for two weeks. Models that attribute sales only to the current period's spend systematically misread which campaigns actually worked.

(iv) **Unquantified Creative Impact.** Two campaigns with identical budgets but different creatives produce radically different results. Standard models treat creative quality as an unquantifiable black box, producing forecasts that are blind to one of the largest drivers of performance variance.

1.4 Contribution of This Work

This paper presents the architecture of a **Brand Simulation Engine**—the first predictive marketing intelligence system designed specifically for the Bangladeshi SME market. The system resolves each of the above failures through a six-layer, graph-augmented simulation stack that combines Bayesian Marketing Mix Modelling, Agent-Based Modelling, Markov Chain attribution, and Genetic Algorithm optimisation, all grounded in a GraphRAG knowledge base powered by Neo4j. The platform is deployed entirely via Next.js on Vercel with full Bangla localisation and aggressive low-bandwidth edge optimisations, making it accessible on the 2G/3G mobile infrastructure that remains prevalent across rural and peri-urban Bangladesh.

2 Theoretical Foundations

2.1 Why MTA Cannot Be the Foundation

Given the market vacuum described in Section 1, one might ask why Bangladeshi businesses cannot simply adopt the attribution tools already used by global brands. The answer lies in the fundamental architectural limitations of Multi-Touch Attribution (MTA).

MTA heuristic models—last-click, first-click, linear—assign conversion credit through predefined, arbitrary rules [1]. They require persistent, individual-level tracking data: the ability to follow a single user from first ad exposure to final purchase across every platform they touched. This requirement has become practically unsatisfiable. The rise of closed "walled-garden" ecosystems means that Meta's conversion data and Google's conversion data are reported separately, with no shared identifier. Privacy legislation has deprecated the third-party cookies that once bridged these silos. The result is a fragmented, incomplete picture that produces attribution outputs that are not merely imprecise—they are systematically biased.

More critically, even a perfectly functioning MTA system would remain *descriptive, not predictive.* It can report what happened; it cannot simulate what *will* happen if the budget is reallocated. It is incapable of answering the questions that matter most before capital is deployed. This is the foundational reason why the architecture presented here abandons MTA entirely, building instead on probabilistic, aggregated modelling frameworks that are both privacy-safe and genuinely forward-looking.

2.2 Marketing Mix Modelling (MMM)

The industry standard for high-level budget simulation is Marketing Mix Modelling—a multi-linear regression methodology that decomposes total sales into two components [1,2]:

- **Base Sales.** Revenue generated organically through long-term brand equity, pricing, and distribution independent of short-term advertising.
- **Incremental Sales.** The direct, causal revenue lift attributable to specific marketing stimuli.

By isolating these components, the engine accurately computes Incremental Return on Ad Spend (iROAS), preventing the erroneous attribution of organic baseline momentum to advertising spend.

2.3 Agent-Based Modelling (ABM)

MMM excels at macroscopic budget allocation but is blind to targeting nuances: a campaign delivering 100,000 impressions to urban millennials is treated identically to one targeting rural demographics. Agent-Based Modelling (ABM) provides the necessary *micro-simulation* layer [1]. Each simulated agent represents a fictive consumer with unique demographic and psychographic attributes (geolocation, brand loyalty, media consumption habits). When a simulated campaign is introduced into the ABM environment, agents react, interact, and generate emergent aggregate phenomena—including viral word-of-mouth diffusion and long-term brand perception shifts.

The comprehensive engine employs a **hybrid approach**: MMM for baseline macroeconomic elasticity, and ABM (implemented via **Mesa 3.0** to leverage high-performance vectorized `AgentSet` operations) for granular audience-specific scenario testing.

2.4 Bayesian Inference

The most critical algorithmic advancement is the transition from frequentist to Bayesian statistics [1]. Marketing data suffers from:

- **Sparsity.** Event-based tactics (seasonal promotions, product launches) occur infrequently.
- **Multicollinearity.** Brands run campaigns across multiple channels simultaneously, making it statistically difficult to isolate individual channel impact.

- **Low Variance.** Marketers optimise for steady returns, limiting the experimental signal available to frequentist methods.

Bayesian MMM incorporates *priors*—historical benchmarks, A/B test results, or industry elasticity estimates—as a foundational layer before analysing new data. The model executes a continuous knowledge-acquisition process: leaning on the prior when data is sparse or collinear, updating it when strong empirical signals emerge. This prevents overfitting and ensures the engine generalises to future conditions.

Table 1: Comparison of Marketing Measurement Paradigms

Paradigm	Core Methodology	Strengths	Weaknesses
MTA	Heuristic rules; bottom-up user tracking via cookies	Granular journey mapping; real-time individual tracking	Vulnerable to privacy law; ignores external factors; arbitrary rules
MMM	Multi-linear regression; Bayesian inference; top-down	Robust ROI forecasting; privacy-compliant; handles macro factors	Can lack micro-level targeting; requires significant history
ABM	Algorithmic simulation of autonomous consumer agents	Models emergent behaviour, word-of-mouth, targeted demographics	Computationally intensive; calibration bias risk

3 Input Parameter Matrix

The predictive validity of the simulation engine is fundamentally determined by the comprehensiveness and quality of its inputs. The system classifies all ingested data into four actionable categories [1, 2].

Table 2: Input Parameter Matrix

Category	Specific Parameters	Function within the Simulation
Endogenous (Controllable)	Ad spend, impressions, CPC, base pricing, discounts, distribution	Primary independent variables marketers manipulate in "what-if" scenarios
Exogenous (Uncontrollable)	Inflation, consumer confidence, seasonality, weather, competitor share of voice	Control variables that establish baseline sales momentum and isolate true marketing lift
Transactional (Outcome)	Revenue, conversions, AOV, CAC, LTV, retention rates	Dependent variables the engine forecasts and optimises against
Audience (Pervasive)	Aggregated intent signals, lifestyle clusters, geolocation segments	Modulators determining how dynamic cohorts respond to specific stimuli

3.1 Competitor Proxy Data via Firecrawl and Crawl4AI

Because direct competitor advertising spend is confidential, the engine relies on proxy data ingested autonomously via a hybrid scraping architecture [2]. **Firecrawl**—a managed web data extraction API designed for AI pipelines—is reserved for complex, heavily protected competitor domains. For routine, high-volume scraping of undefended targets, the system utilizes **Crawl4AI**. Both tools extract JavaScript-rendered pages, stripping boilerplate to produce noise-free Markdown suitable for vector embedding. Competitor pricing shifts, feature launches, and promotional density updates are ingested on a scheduled basis, continuously updating the penalty parameters within the simulation.

4 Mathematical Transformations of Consumer Behaviour

Raw historical data cannot be fed directly into a regression model; it must first be transformed to reflect the psychological and economic realities of consumer behaviour [1].

4.1 Adstock Transformation: Modelling Temporal Lag

Marketing stimuli rarely produce instantaneous conversions. A consumer may encounter a streaming advertisement on Monday but delay purchase until Saturday. The **Adstock transformation** models the memory retention, build-up, and eventual decay of advertising exposure via an autoregressive process:

$$A_t = X_t + \lambda\, A_{t-1} \tag{1}$$

where A_t is the transformed Adstock value at time t, X_t is the raw media spend or exposure at time t, and $\lambda \in [0, 1]$ is the *decay rate* (retention parameter). A television campaign with $\lambda = 0.8$ carries 80% of its psychological impact into the subsequent week. Different channels exhibit distinct decay rates: lower-funnel paid search ads decay almost immediately ($\lambda \approx 0.1$–0.2), while top-of-funnel brand-building TV campaigns sustain long residual tails ($\lambda \approx 0.7$–0.9).

By applying the Adstock transformation to historical inputs before regression, the engine accurately attributes delayed sales to the correct historical spend—enabling realistic simulations of always-on versus flighted (intermittent) campaign scheduling.

4.2 The Hill Function: Modelling Diminishing Returns

A catastrophic error in rudimentary forecasting is the assumption of linear scalability. If \$10,000 yields 100 conversions, a naïve model predicts \$1,000,000 yields 10,000 conversions. Modern digital marketing is governed by *diminishing returns*: as campaigns scale, the algorithm exhausts high-intent audiences and begins serving impressions to progressively less relevant users, driving up the cost-per-acquisition.

The engine enforces this economic reality through the **Hill Function**, which transforms the linear Adstocked spend into a non-linear S-curve:

$$f(x) = \frac{x^S}{K^S + x^S} \tag{2}$$

where x is the Adstocked media variable (effective spend), S is the *shape parameter* (the slope, determining how rapidly the response accelerates), and K is the *half-saturation parameter* (the exact spend level at which the channel achieves 50% of its maximum potential efficiency). The engine extracts unique (S, K) pairs for every individual marketing channel during the training phase. When a marketer simulates a budget increase for a specific channel, the system processes the new budget through that channel's Hill function—predicting precisely when an investment crosses from profitable scalability into wasted capital.

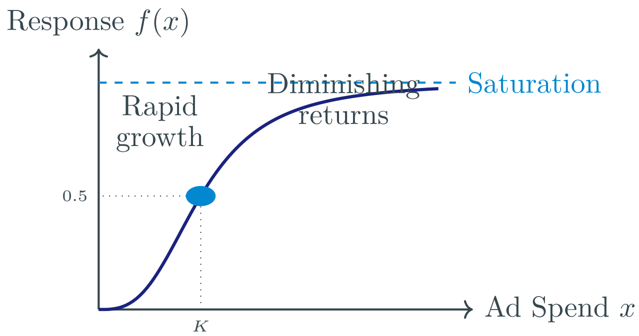

Figure 1: Hill Function S-curve. The parameter K denotes the half-saturation point at which 50% of maximum channel efficiency is achieved. Spend beyond this inflection enters the region of diminishing returns.

4.3 Markov Chain Micro-Funnel Simulation

While Bayesian MMM optimises macro-budgets, marketers also require insight into the micro-dynamics of the customer journey. **Markov Chain modelling** provides a stochastic framework in which each marketing touchpoint (Organic Search, Social Ad, Email, Retargeting) represents a distinct "state". The model analyses historical customer journey logs to construct a *transition probability matrix* **P**, where each entry P_{ij} denotes the probability of a consumer moving from state i to state j:

$$\mathbf{P} = \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{pmatrix}, \quad \sum_j p_{ij} = 1 \;\; \forall\, i \tag{3}$$

The true incremental value of any channel is determined via the **Removal Effect**: the engine computes baseline conversion probability, removes a specific channel node from the graph, recomputes the conversion probability, and quantifies the delta as that channel's causal contribution. This allows the simulation to predict the cascading secondary effects of budget cuts—for instance, how reducing top-of-funnel display spend will starve bottom-of-funnel branded search, ultimately degrading final conversions.

5 Solving the Cold-Start Problem

A critical vulnerability in static predictive models is the absence of historical data for entirely novel campaigns, new geographic markets, or unprecedented product categories. Traditional regression algorithms fail catastrophically when forced to predict outcomes for entities lacking interaction logs—a challenge known as the **Cold-Start Problem** [1].

5.1 Transfer Learning and Meta-Learning

Rather than attempting to train from scratch on empty data, the engine employs deep neural networks capable of **Transfer Learning**. These networks are pre-trained on large cross-industry repositories of historical marketing data, learning the underlying physics of consumer demand, price sensitivity, and channel efficacy. When a new "cold" campaign is introduced, the system applies inverse propensity weighting and multimodal data embedding to identify the closest historical analogues. Performance benchmarks, baseline elasticity, and saturation curves from analogous past campaigns are transferred to initialise the simulation.

Meta-learning architectures further allow the neural network to rapidly adjust its internal weights based on the very first trickle of real-time performance data, accelerating correction of cold-start estimates in near real time.

5.2 Synthetic Data Generation

In scenarios where historical analogues are insufficient or restricted by privacy regulations, the engine relies on **Synthetic Data Generation**. Generative AI and procedural algorithms manufacture vast datasets that preserve the statistical properties, correlation matrices, and distributions of real-world consumer behaviour without containing traceable customer information. Frameworks such as RetailSynth utilise discrete choice models and utility theory to simulate hundreds of thousands of synthetic shopping transactions, capturing heterogeneous price sensitivities and promotional responses [1].

6 System Architecture

The engine is structured as a six-layer, event-driven stack. Each layer is independently scalable and communicates via well-defined APIs and message queues, ensuring the computationally intensive simulation layer remains fully decoupled from the user-facing application.

6.1 Layer 1 – Data Sources

Three classes of data feed the ingestion pipeline: **Endogenous** inputs (ad platforms, CRM, promotional data), **Exogenous** inputs (competitor share of voice scraped via Firecrawl, macroeconomic time-series), and **Transactional** inputs (historical revenue logs, conversion events, LTV).

6.2 Layer 2 – Ingestion & Processing

Batch data is pulled securely by ETL tools (Airbyte or Fivetran). Real-time scraping events and streaming telemetry flow through a message queue (Kafka or Redis Pub/Sub), decoupling ingestion from downstream processing to prevent back-pressure.

6.3 Layer 3 – Storage & Knowledge Base

- **Neo4j (Graph DB).** Serves as the primary relationship mapper, storing the marketing input matrix as interconnected nodes and edges (e.g., `Campaign -[TARGETS]-> AudienceSegment`).
- **Pinecone / Weaviate (Vector Store).** Stores embeddings of creative assets and textual campaign data, enabling semantic similarity search for the GraphRAG retrieval pipeline.
- **BigQuery / Snowflake (Data Lake).** Cold-storage repository for raw, unstructured historical logs processed in batch.

6.4 Layer 4 – AI & Simulation Engine

Heavy mathematical simulations execute in a dedicated Python environment (FastAPI), decoupled from the frontend by a Celery task queue backed by Redis:

- **Macro-Simulation.** Bayesian MMM implemented via PyMC-Marketing performs top-down budget forecasting with Adstock and Hill transformations.
- **Micro-Simulation.** ABM (powered by Mesa 3.0) and Markov Chains simulate granular user journeys and transition probabilities.

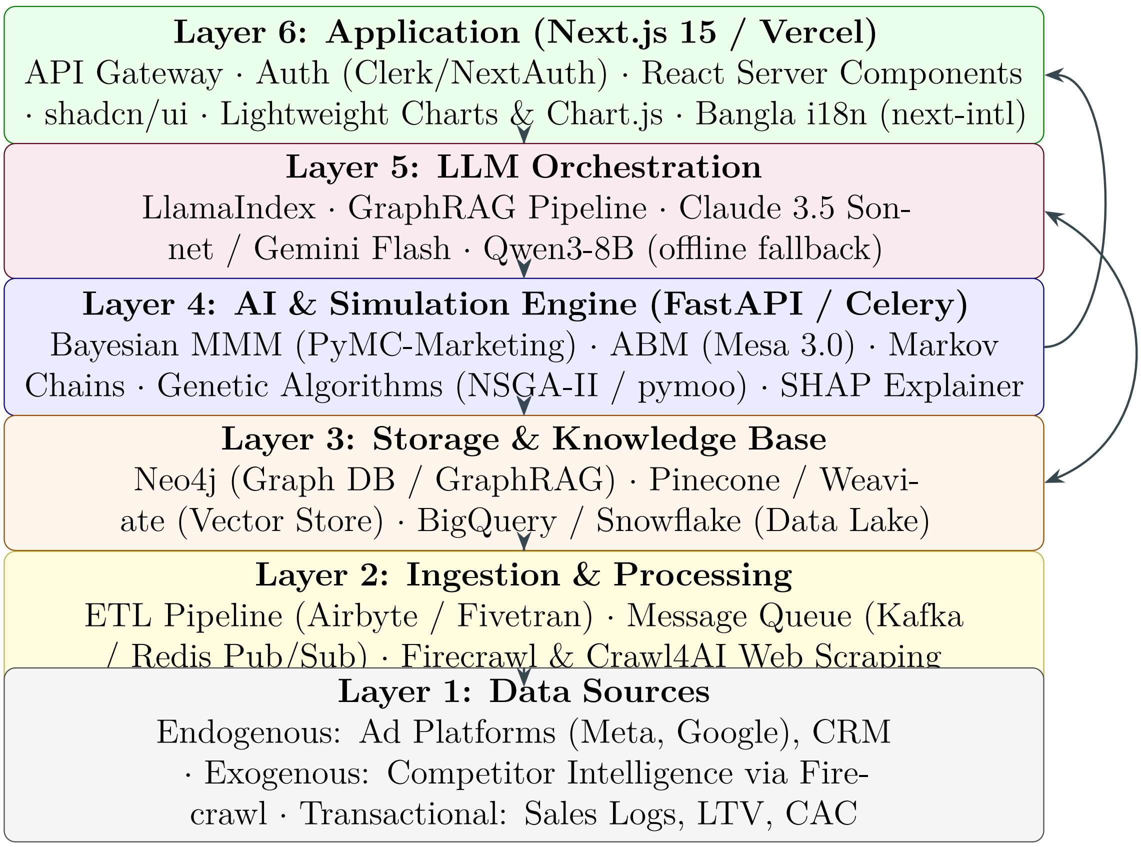

Figure 2: Six-layer system architecture of the Brand Simulation Engine. Bidirectional arrows between the Storage, AI, and LLM layers indicate continuous read/write cycles during simulation execution.

- **Optimization Engine.** NSGA-II Genetic Algorithm (pymoo) evaluates thousands of budget combinations to find the Pareto frontier.
- **SHAP Explainability.** A SHAP TreeExplainer computes the exact mathematical contribution of every input feature before passing outputs to the LLM, guaranteeing deterministic, hallucination-free explanations.

6.5 Layer 5 – LLM Orchestration

The LLM layer bridges raw simulation outputs and natural-language explainability via the GraphRAG pipeline (Section 7) orchestrated by **LlamaIndex**. Cloud inference uses Claude 3.5 Sonnet or Gemini 1.5 Flash via the Vercel AI SDK. An offline fallback deploys **Qwen3-8B** locally via Ollama, with a *Fast Profile* that suppresses the internal reasoning chain for routine explanations and a *Deep Profile* reserved for complex scenario ideation.

6.6 Layer 6 – Application Layer

Next.js 15 with React Server Components (RSC) renders heavy data server-side, keeping the client bundle lean. Authentication is managed by Clerk or NextAuth. Visualisations are built with shadcn/ui, **Lightweight Charts**, and **Chart.js**. Full Bangla i18n is implemented via next-intl with namespaced JSON dictionaries and locale-based sub-path routing.

7 GraphRAG: Graph-Powered Retrieval-Augmented Generation

Standard RAG architectures rely on vector similarity search, retrieving text chunks based on embedding proximity. While effective for simple factual queries, vanilla RAG fails on the multi-hop reasoning required for marketing simulation [2]. A query such as "Which campaigns targeting urban millennials overlapped with a competitor's promotional cycle and achieved a CAC below target?" requires traversing multiple relationship hops—impossible for pure vector retrieval.

GraphRAG weaves a structured Neo4j knowledge graph into the retrieval pipeline, enabling the LLM to reason across relationships and produce grounded, auditable answers.

7.1 Knowledge Graph Schema

Table 3: Neo4j Knowledge Graph Schema

Component	Type	Function
Nodes	Campaign, Channel, Product, AgentCluster, Competitor, Outcome	Core entities within the simulation environment
Properties	Ad_Spend, Impressions, Price, Decay_Rate, Saturation_Point, Inflation_Rate	Quantitative attributes enabling mathematical transformations
Causal Edges	INFLUENCES, SUPPRESSES, CANNIBALIZES, GENERATES	Defines relationships (e.g., Competitor SUPPRESSES Campaign)
Structural Edges	ALLOCATED_TO, TARGETS, BELONGS_TO, INTERACTS_WITH	Maps the hierarchy of the marketing mix

7.2 Indexing and Community Detection

The indexing phase transforms unstructured campaign reports and structured transaction logs into the Neo4j knowledge graph via **LlamaIndex**. Community detection algorithms then cluster data hierarchically into semantic groups, and the LLM pre-generates summaries for each community, enabling both granular local search (entity neighbourhood traversal) and broad global search (community-level reasoning) without traversing the entire graph on every query [2].

7.3 Hybrid Retrieval at Query Time

A custom retriever combines vector similarity results with k-hop neighbourhood traversal, passing the synthesised context window to the LLM. A text-to-Cypher translation service allows the LLM to generate native graph queries dynamically. The resulting answer is not only factually grounded but carries explicit provenance—the marketer can audit each recommendation against its source nodes in the graph.

8 Prescriptive Optimisation Engine

Once the engine has modelled temporal lag, established non-linear saturation curves, and integrated creative quality coefficients, it must prescribe the mathematically optimal resource allocation across channels.

8.1 Genetic Algorithms (NSGA-II)

The global marketing optimisation problem is a multi-objective, non-linear, heavily constrained search through a high-dimensional space. Genetic Algorithms simulate natural selection to iteratively discover optimal solutions [1]:

Step 1. Initialisation. A large random population of budget allocations ("chromosomes") is generated across all channels.

Step 2. Evaluation. Each allocation is fed into the Bayesian MMM and Hill function model to compute its "fitness" (projected revenue and ROI).

Step 3. Selection. The highest-performing allocations are selected for "reproduction"; sub-optimal ones are discarded.

Step 4. Crossover & Mutation. Successful allocations are mathematically combined (crossover), and random variations (mutations) are introduced to prevent entrapment in local optima.

Implemented via the `pymoo` library (NSGA-II algorithm), the engine iterates thousands of generations in seconds. The output is a **Pareto frontier**—a mathematically verified set of budget allocations where no single channel's performance can be improved without degrading another [2].

8.2 SHAP Deterministic Explainability

Before the LLM generates any narrative report or recommendation, all simulation outputs are passed through a **SHAP (SHapley Additive Explanations) TreeExplainer** [3]. SHAP computes the exact marginal contribution of every input feature (budget level, audience segment, platform, competitor index) to the final prediction. This guarantees that LLM-generated explanations are mathematically anchored to the model's actual behaviour, eliminating hallucinated justifications.

9 Bangladesh-Specific NLP and Localisation

For the simulation engine to achieve mass adoption among Bangladeshi SME marketers, the platform must natively understand the linguistic realities of the local market—including code-mixed "Banglish" social media text that standard multilingual models handle poorly [2, 3].

9.1 Bangla-Native NLP Pipeline

- **`csebuetnlp/banglabert`.** An ELECTRA-architecture transformer trained with a Replaced Token Detection objective, making it exceptionally powerful at natively processing noisy, code-mixed Bengali and English social media sentiment. Used for sentiment analysis and creative feature extraction from Bangla ad copy.
- **`BAAI/bge-m3`.** A multi-vector embedding model supporting dense, sparse (keyword), and multi-vector retrieval across 100+ languages. It preserves the semantic meaning

of Banglish queries without collapsing local context, feeding clean, culturally aware embeddings into the Pinecone/Weaviate vector store.

9.2 Next.js i18n Architecture

The application implements locale-based sub-path routing (`domain.com/bn/dashboard` for Bangla, `domain.com/en/dashboard` for English), providing SEO-indexable, locale-specific URLs. Translations are managed with **next-intl**, loading namespace-scoped JSON dictionaries (e.g., `bn/finance.json`, `bn/navigation.json`) exclusively server-side via React Server Components. TypeScript types enforce compile-time key synchronisation, preventing silent translation divergence [2].

9.3 Low-Bandwidth Edge Optimisation

Rural and peri-urban Bangladesh frequently operates on 2G/3G mobile connections with high latency and packet loss. The deployment applies four layers of mitigation:

(1) **Font Subsetting.** The `next/font` module self-hosts fonts and generates subset files containing only the Unicode ranges required for Bangla, reducing font payloads from ∼600 KB to under 50 KB.

(2) **Incremental Static Regeneration (ISR).** Standard reporting views and historical Pareto frontiers are pre-computed and cached on Vercel's global Edge Network, serving requests from geographically proximate nodes in under 100 milliseconds.

(3) **Selective Prefetching.** The default aggressive Next.js `<Link>` prefetching is disabled for non-critical routes. Prefetch triggers are bound to explicit user intent (hover / touch), preventing wasteful background data transfers on metered mobile connections.

(4) **Dynamic Imports.** Heavy visualisation libraries such as Lightweight Charts and Chart.js are loaded via `next/dynamic` with lazy evaluation, minimising the initial Time to Interactive (TTI) on low-end devices.

9.4 Offline / Rural Fallback Mode

For users with intermittent connectivity, the architecture provides an offline-capable local LLM profile. **Qwen3-8B** is deployed via **Ollama** on the user's device. A *Fast Profile* disables the model's internal reasoning chain (`<|think|>` token suppression) for routine explanations, ensuring rapid response times on low-power hardware. A *Deep Profile* enables full reasoning for complex scenario ideation when resources permit [3].

10 Executive Dashboard and Decision-Support Metrics

The complex mathematics of Bayesian inference, Markov Chains, and Genetic Algorithms must be abstracted into an intuitive, forward-looking interface for marketing executives and CFOs [1, 2].

10.1 Component Architecture

The dashboard is built on **Next.js 15 RSC** and **shadcn/ui**—a collection of re-usable components copied directly into source code (rather than installed as an opaque NPM dependency), granting full ownership and deep customisation. A persistent sidebar transitions to a Sheet drawer on mobile. TanStack Table powers sortable, filterable `DataTable` components handling tens of thousands of transactional rows without degrading performance.

10.2 Visualisation Components

Table 4: Dashboard Visualisation Components

Component	Rendering Engine	Marketing Function
Saturation S-Curve	Lightweight Charts (Canvas)	Visualises the Hill function per channel; delineates growth from diminishing returns
Budget Allocation	Chart.js (Canvas)	Displays capital distribution from the Genetic Algorithm's Pareto-optimal output
ROI & iROAS Tracking	Lightweight Charts	Tracks incremental return over time; proves causal advertising impact
Micro-Funnel Journey	Multi-node graph	Represents Markov Chain transition probabilities between touchpoints

10.3 Key Decision-Support Metrics

Table 5: Financial Decision-Support Metrics Output by the Engine

Metric	Financial Definition	Strategic Value
ROI	Total revenue generated relative to total cost	Macro-level profitability view; justifies overall budget existence
iROAS	Revenue directly caused by advertising, net of baseline sales	Proves true causal marketing impact; satisfies CFO incrementality demands
mROI	Projected revenue from the *next* incremental dollar spent	Identifies diminishing-return thresholds; dictates when to stop funding a channel
Probability Distributions	Forecasts expressed as confidence intervals	Replaces point-estimate guesswork with rigorous financial risk quantification

11 Agentic Development Workflow

The velocity required to deliver a production-ready full-stack marketing application within BuildFest timelines demands an AI-native development paradigm [2, 3].

Claude Code (Anthropic) operates as a terminal-native autonomous agent. Provided a high-level natural language prompt describing the application schema, API routes, and database connections, it scaffolds the Next.js App Router, connects Neo4j and Firecrawl layers, and manages git operations autonomously. The Vercel MCP plugin dynamically injects a knowledge graph of the entire Vercel ecosystem into the agent's context window, catching deprecated patterns via `postToolUse` validation hooks and enabling autonomous deployment commands (`/vercel-plugin:deploy prod`).

Cursor IDE, an AI-native editor built on VS Code, then takes over for granular human-in-the-loop refinement: tweaking UI components, perfecting dashboard styling, and refining complex state logic. This sequential pipeline—Claude Code for broad scaffolding, Cursor for visual polish—condenses development timelines from months to hours.

12 Conclusion

This paper has presented a comprehensive technical architecture for a Graph-Augmented Bayesian Simulation Engine purpose-built for SME marketers in Bangladesh. The system addresses the fundamental deficiencies of legacy Multi-Touch Attribution by replacing heuristic, correlation-based credit assignment with causally rigorous, mathematically grounded pre-launch simulation.

The core scientific contribution is the hybrid triple-engine simulation core: **Bayesian MMM** with Adstock and Hill transformations provides macroscopic, econometrically sound budget forecasting; **Agent-Based Modelling** captures granular audience-level dynamics invisible to aggregated regression; and **Markov Chain** attribution quantifies the removal effect of individual channels across the conversion funnel. All outputs are globally optimised by an NSGA-II Genetic Algorithm and rendered interpretable by SHAP explainability guarantees before the LLM synthesises the final narrative.

The GraphRAG backend, powered by Neo4j, elevates retrieval-augmented generation beyond vector similarity, enabling multi-hop causal reasoning that eliminates hallucinated financial recommendations. A hybrid ingestion architecture utilizing Firecrawl and Crawl4AI continuously calibrates exogenous penalty parameters against live competitor intelligence, resolving the cold-start problem without dependence on scarce historical data.

Critically, the architecture is not merely theoretically sound—it is engineered for the physical realities of the Bangladeshi market. Font subsetting, ISR edge caching, selective prefetching, and an offline Ollama-powered LLM fallback ensure the platform remains fully functional on 2G/3G mobile networks. Native Bangla NLP via `banglabert` and `bge-m3` makes the system culturally legible to local SME operators, removing the language barrier that historically excluded them from advanced analytics tools.

The resulting platform transitions the marketing function from a reactive cost centre reliant on retrospective attribution into a proactive revenue driver powered by prescriptive simulation—delivering, for the first time, enterprise-grade predictive marketing intelligence to the SMEs of an emerging economy.

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